

C++ / Program Analysis Engineer

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I build analyzers and reusable systems components in C++ and C. The primary evidence is the conservative PS-form Analyzer with an exact oracle, 12 AdvancedAlgorithms modules, IR/SSA optimizations in Wist2 and an educational x86-64 codegen/register-allocation laboratory.

1st out of 49
the only 104/104 result in the MCST selection

C++23 · strict builds
warnings-as-errors, ASan/UBSan, clang-tidy

Exact oracle + metamorphic tests
WA/TL hunting and counterexample retention

EXPERIENCE

- July–August 2026

MCST — Compiler Engineering Intern
LLVM track · C++23 · remote
 - Built a shared directed-graph core and four components: RPO with cycle detection, Dijkstra, Tarjan SCC, and Dinic max-flow.
 - Defined graph-input and unknown-node error contracts; handled self-loops, parallel edges, unreachable vertices, and disconnected components.
 - Configured C++23/CMake builds with warnings-as-errors, ASan/UBSan, clang-tidy, and automated I/O tests.
- 2024 — present

UniversalToolchain / Wist2 — Creator and Primary Developer
Compiler/runtime architecture · verification engineering
 - Designed multiple IR layers, two execution paths, and an opt-in AIR → SSA → AIR route.
 - Added capability contracts, structural verifiers, and interpreter/CIL parity tests.
- 2021 — present

Programming Tutor
C/C++ · Python · algorithms
 - Taught about 50 students in C/C++, Python, and algorithms; regularly review code.

SKILLS

RECOGNITION

- Programming olympiads**
Prize-winner of the HSE “Vysshaya Proba” Olympiad in competitive and industrial programming; prize-winner of the regional Russian Informatics Olympiad.
- Compiler projects**
Two highest-tier diplomas in the MEPhI Junior competition and a highest-tier diploma in the Baltic Science and Engineering Competition.

EDUCATION

HSE University student

SELECTED PROJECTS

- C17 · Python · program analysis

PS-form Memory Dependence Analyzer
A conservative cross-iteration memory-dependence analyzer: yes and no require proof; unresolved cases remain maybe .
 - 1st out of 49, the only 5.0/5.0 and 104/104 result.
 - normalization, range/residue/GCD filters, exact affine analysis, monotonic scan, and cost-bounded enumeration.
- C++23 · header-only library

AdvancedAlgorithms
Reusable algorithms and data structures with explicit contracts, reference oracles and large-input tests.
 - Centroid decomposition, HLD, LCA, Dinic, iterative Tarjan, bridges/articulation points and Dijkstra.
 - DSU, lazy segment tree, sparse table, Aho–Corasick and simulated annealing.
- Rust · x86-64 · register allocation

Codegen & Register Allocation Playground
An educational backend lab with a reference interpreter and isolated native-code execution.
 - SSA-like IR, liveness, live intervals, and linear scan.
 - x86-64 emission and disassembly through iced-x86.